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Course: Compiler Design

Assignment Lab-2

Task 1:

Write a C++ program that takes a full line as input and prints all operators found in order, then prints the total count. Consider common operators such as + - * / = % < > ! & | (treat each character operator separately for this lab).

Sample Input:

2+3=5

Expected Output:

operator1 : +

operator2 : =

number of operators = 2

Answer:

```
#include <iostream>
#include <string>

using namespace std;
int main() {
    string input;
    string target = " ,+ ,/ ,* ,^ ,$, < ,> ,! ,=";
    int count = 0;
    cout << "Enter an expression :";
    getline(cin, input);

    for (int i = 0; i < input.length(); i++) {
        for (int j = 0; j < target.length(); j++) {
            if (input[i] == target[j]) {
                count++;
                cout << "operator" << count << " : " << input[i] << endl;
                break;
            }
        }
    }
    cout << "Number of operators = " << count << endl;
    return 0;
}
```

Output:

```
"E:\Downloads\Compiler Mid  ×  +  v
Enter an expression :3*8+6/7=7
operator1 : *
operator2 : +
operator3 : /
operator4 : =
Number of operators = 4

Process returned 0 (0x0)   execution time : 25.236 s
Press any key to continue.
```

Task 2:

Write a C++ program that takes a full line as input and determines whether it is

1. a single line comment,
2. a multi line comment, or
3. not a comment.

Rules:

- Single line comment starts with //
- Multi line comment starts with /* and ends with */

Constraint: do not use more than **two** if statements in your solution.

Sample Input:

```
/*abc*/
```

Expected Output:

This is a multi line comment.

Answer:

```
#include <iostream>
#include <string>

using namespace std;
int main() {
    string input;
    cout << "Enter text: ";
    getline(cin, input);

    if (input.substr(0, 2) == "//") {
        cout << "This is a single line comment." << endl;
    }
    else if (input.length() >= 4 && input.substr(0, 2) == "/*" && input.substr(input.length() - 2) == "*/") {
```

```

    cout << "This is a multi line comment." << endl;
}
else {
    cout << "Not a comment." << endl;
}

return 0;
}

```

Output:

The image shows two screenshots of a C++ compiler window titled "E:\Downloads\Compiler Mid".

The first screenshot shows the output for a multi-line comment:


```

Enter text: /*abc*/
This is a multi line comment.

Process returned 0 (0x0)   execution time : 10.670 s
Press any key to continue.
    
```

The second screenshot shows the output for a single-line comment:


```

Enter text: //this is my work
This is a single line comment.

Process returned 0 (0x0)   execution time : 7.931 s
Press any key to continue.
    
```

Task 3:

Write a C++ program that takes a single token (no spaces) and prints:

- Keyword if it matches one of these keywords: int, float, double, char, if, else, while, return
- Identifier if it is a valid identifier (starts with letter or underscore, then letters, digits, underscores only)
- Invalid otherwise

Sample Input:

while

Expected Output:

Keyword

Answer:

```
#include <iostream>
```

```

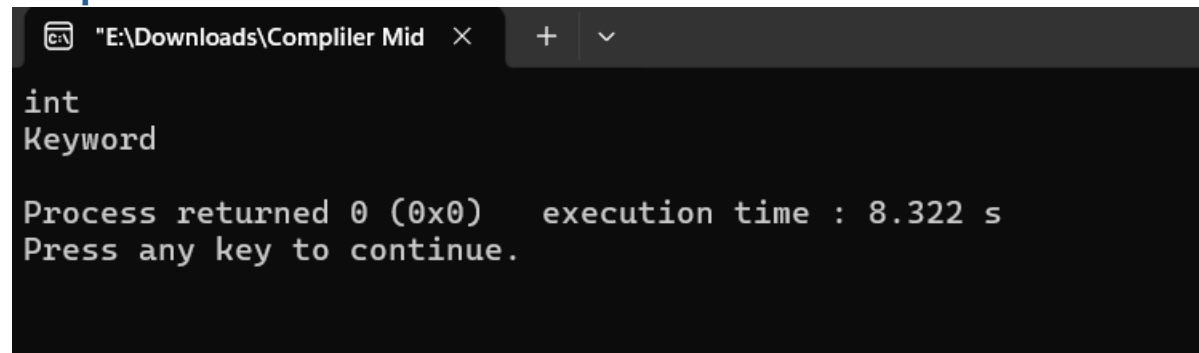
#include <string>
using namespace std;

int main() {
    string token;
    cin >> token;
    if (token == "int" || token == "float" || token == "double" ||
        token == "char" || token == "if" || token == "else" ||
        token == "while" || token == "return") {
        cout << "Keyword" << endl;
    }
    else if (isalpha(token[0]) || token[0] == '_') {
        bool valid = true;
        for (int i = 1; i < token.length(); i++) {
            if (!isalnum(token[i]) && token[i] != '_') {
                valid = false;
                break;
            }
        }
        if (valid)
            cout << "Identifier" << endl;
        else
            cout << "Invalid" << endl;
    }
    else {
        cout << "Invalid" << endl;
    }

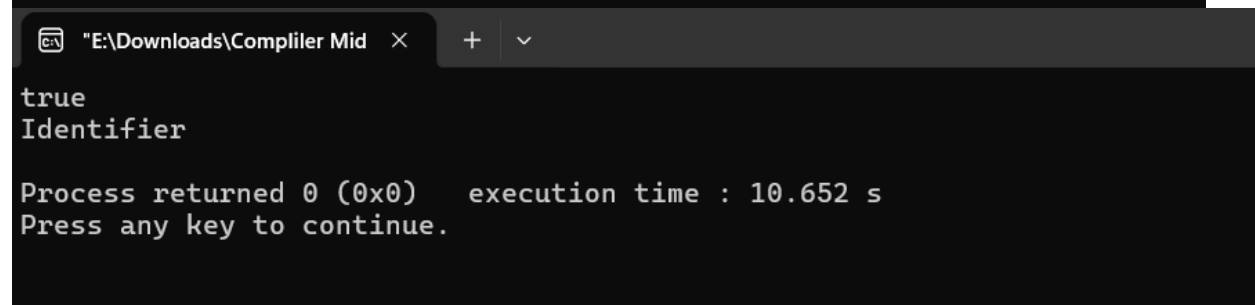
    return 0;
}

```

Output:



A screenshot of a Windows command prompt window. The title bar shows the file path "E:\Downloads\Compiler Mid" and standard window controls. The command prompt displays the output of the program: "int" followed by "Keyword" on the next line. Below this, it shows "Process returned 0 (0x0) execution time : 8.322 s" and "Press any key to continue.".



A screenshot of a Windows command prompt window, similar to the one above. The title bar shows the file path "E:\Downloads\Compiler Mid". The command prompt displays the output of the program: "true" followed by "Identifier" on the next line. Below this, it shows "Process returned 0 (0x0) execution time : 10.652 s" and "Press any key to continue.".

```
"E:\Downloads\Compiler Mid  ×  +  ∨  
;  
Invalid  
  
Process returned 0 (0x0)   execution time : 2.852 s  
Press any key to continue.  
|
```

Task 4:

Write a C++ program that takes a single token and prints:

- Integer literal if it contains only digits, optionally preceded by + or -
- Floating literal if it contains digits with exactly one decimal point, optionally preceded by + or -, and at least one digit on both sides of the decimal point
- Invalid number otherwise

Sample Input:

-12.50

Expected Output:

Floating literal

Answer:

```
#include <iostream>  
using namespace std;  
  
int main() {  
    float num;  
  
    cout << "Enter a number: ";  
    cin >> num;  
  
    if (cin.fail()) {  
        cout << "invalid" << endl;  
    }  
    else if (num == (int)num) {  
        cout << "integer integral" << endl;  
    }  
    else {  
        cout << "floting integral" << endl;  
    }  
  
    return 0;  
}
```

Output:

```
"E:\Downloads\Compiler Mid" x + v
Enter a number: -44
integer integral

Process returned 0 (0x0)   execution time : 12.384 s
Press any key to continue.

"E:\Downloads\Compiler Mid" x + v
Enter a number: 78.99
floting integral

Process returned 0 (0x0)   execution time : 4.824 s
Press any key to continue.

"E:\Downloads\Compiler Mid" x + v
Enter a number: a
invalid

Process returned 0 (0x0)   execution time : 3.584 s
Press any key to continue.
```

Task 5:

Write a C++ program that takes a full line as input, prints each delimiter in order, then prints the total count. Use these delimiters:

(){}[] , ;

Sample Input:

```
int f(int a, int b){ return a+b; }
```

Expected Output:

delimiter1 : (

delimiter2 : (

delimiter3 : ,

delimiter4 :)

delimiter5 :)

delimiter6 : {

delimiter7 : ;

```
delimiter8 : }  
number of delimiters = 8
```

Answer:

```
#include <iostream>  
  
#include <string>  
  
using namespace std;  
  
int main() {  
    string line;  
  
    getline(cin, line);  
  
    int count = 0;  
  
    for (int i = 0; i < line.length(); i++) {  
        char c = line[i];  
  
        if (c == '(' || c == ')' || c == '{' || c == '}' ||  
            c == '[' || c == ']' || c == ';' || c == ',') {  
            count++;  
  
            cout << "delimiter" << count << " : " << c << endl;  
        }  
    }  
  
    cout << "Number of delimiters = " << count << endl;  
  
    return 0;  
}
```

Output:


```
int f(int a, int b){ return a+b; }
delimiter1 : (
delimiter2 : ,
delimiter3 : )
delimiter4 : {
delimiter5 : ;
delimiter6 : }
number of delimiters = 6

Process returned 0 (0x0)    execution time : 45.700 s
Press any key to continue.
```

////////////////////////////////// End //